

Mass versus Velocity at the Characteristic Time: $N = 4, 8, 16$

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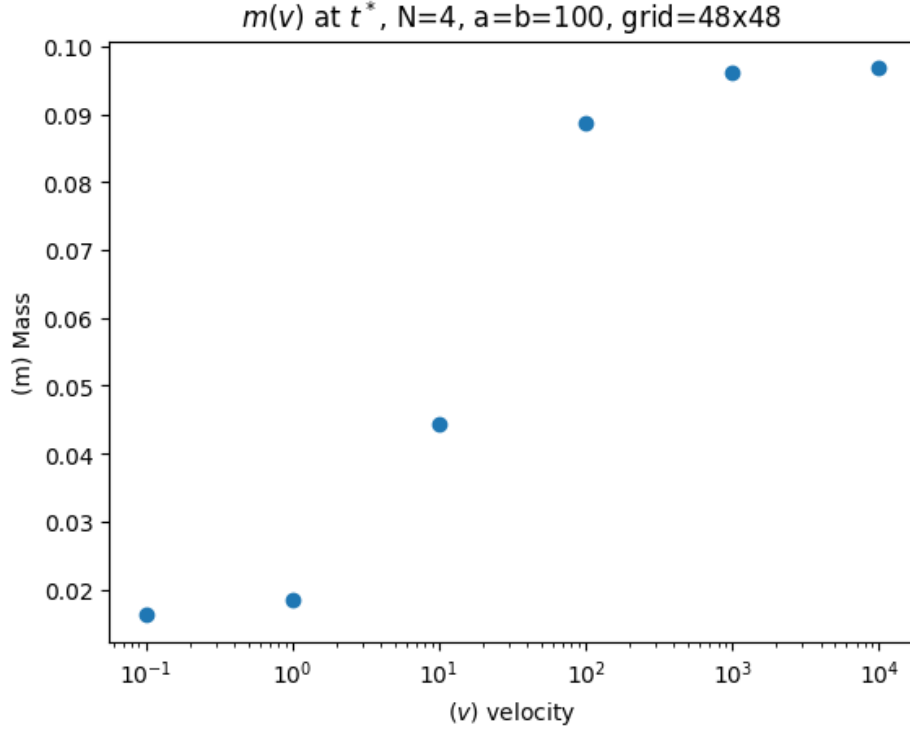


Figure 1: Total mass as a function of filament velocity, $m(v)$, for $N = 4$. Each point is captured at the characteristic time t^* of its own run, not at a common time.

v	t^*	m^*
10^{-1}	0.0853	0.0162
10^0	0.0845	0.0186
10^1	0.0775	0.0443
10^2	0.0709	0.0888
10^3	0.0702	0.0961
10^4	0.0701	0.0968

Table 1: Characteristic time t^* and the mass m^* captured at that respective t^* , per velocity v , for $N = 4$ (3 significant figures).

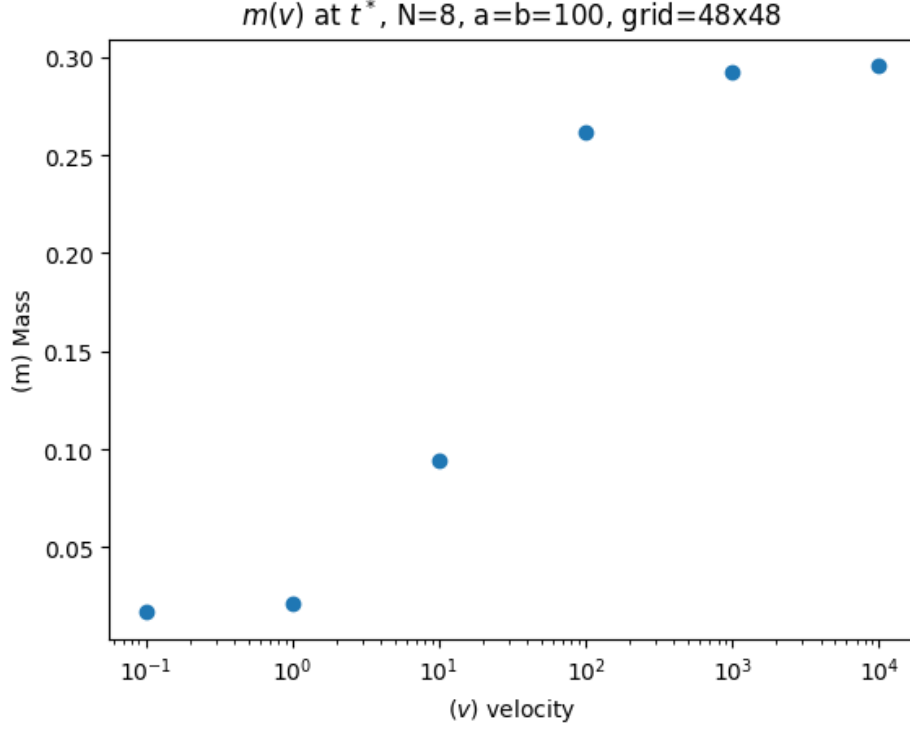


Figure 2: Total mass as a function of filament velocity, $m(v)$, for $N = 8$. Each point is captured at the characteristic time t^* of its own run, not at a common time.

v	t^*	m^*
10^{-1}	0.0914	0.0169
10^0	0.0901	0.0216
10^1	0.0771	0.0941
10^2	0.0636	0.262
10^3	0.0620	0.292
10^4	0.0618	0.295

Table 2: Characteristic time t^* and the mass m^* captured at that respective t^* , per velocity v , for $N = 8$ (3 significant figures).

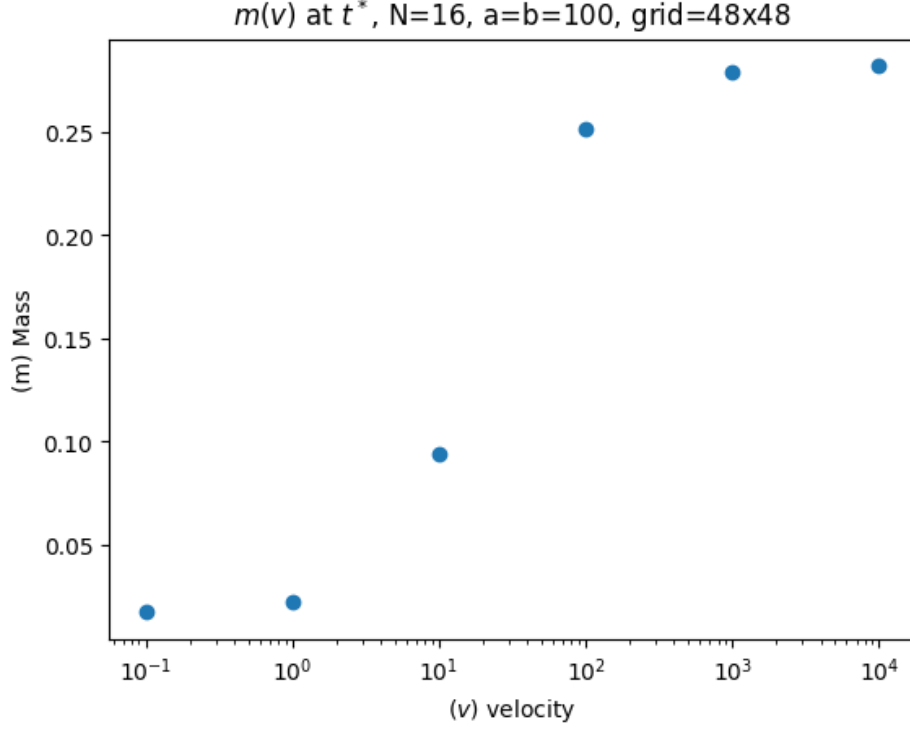


Figure 3: Total mass as a function of filament velocity, $m(v)$, for $N = 16$. Each point is captured at the characteristic time t^* of its own run, not at a common time.

v	t^*	m^*
10^{-1}	0.0913	0.0170
10^0	0.0900	0.0218
10^1	0.0770	0.0937
10^2	0.0642	0.251
10^3	0.0627	0.279
10^4	0.0625	0.282

Table 3: Characteristic time t^* and the mass m^* captured at that respective t^* , per velocity v , for $N = 16$ (3 significant figures).

A scaling factor in m^* relating $N = 4$ to $N = 8$ and $N = 16$

All three sweeps were run at the same coupling ($a = b = 100$), the same grid (48×48), and over the same six velocities, so the three $m^*(v)$ trajectories can be placed on a common plot by a single constant per N .

Take the mass at the largest velocity of the sweep, $m_N^*(10^4)$, as the reference level. The scaling factor is its ratio to the $N = 4$ value,

$$c_N = \frac{m_N^*(10^4)}{m_4^*(10^4)} \quad (1)$$

and each trajectory is brought onto the common curve by dividing through by it,

$$\hat{m}_N^*(v) = \frac{m_N^*(v)}{c_N}. \quad (2)$$

N	$m_N^*(10^4)$	c_N
4	0.09679	1 (reference)
8	0.29547	3.053
16	0.28216	2.915

Table 4: Mass at $v = 10^4$ and the resulting scaling factor of Eq. (1).

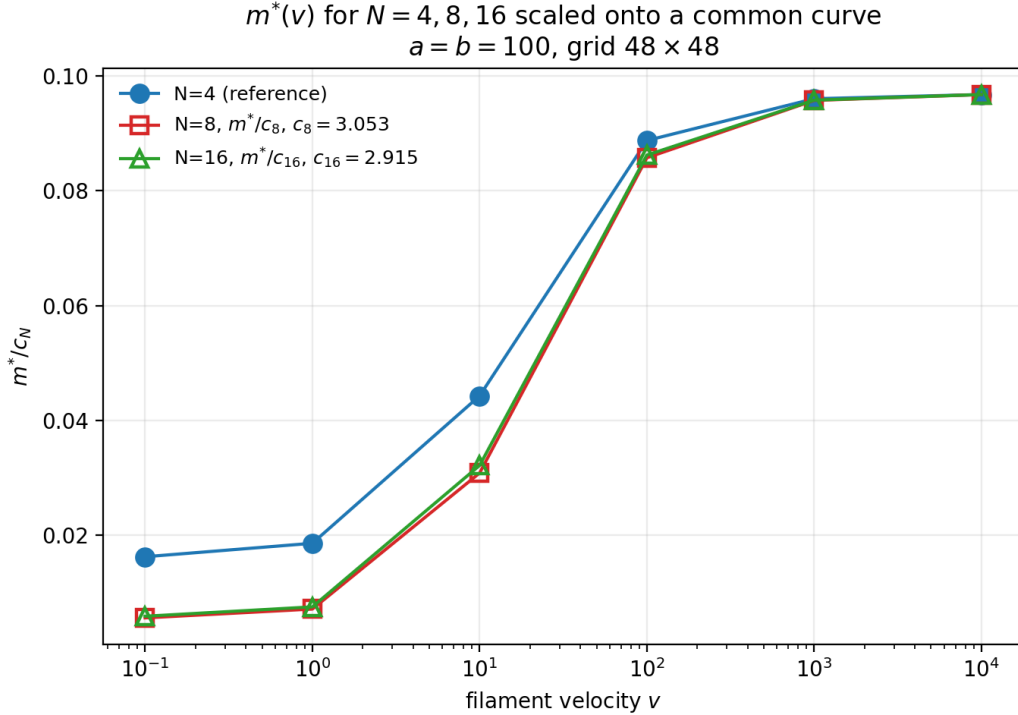


Figure 4: The three trajectories after applying Eq. (2), with the constants c_N of Table 4. The curves lie together over the upper three decades of v ; $N = 8$ and $N = 16$ coincide throughout. The separation at $v \lesssim 10^0$ is expected, since a single multiplicative constant fixed on the plateau cannot also match the low-velocity end.